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NATIONAL INSTITUTE OF TECHNOLOGY,
KURUKSHETRA-136119

B. Tech. (RA) – 4th Sem.; Mid-SEM Exam – II
April - 2026

Roll No.

RAPC-212

Time: 50 Minutes

Manufacturing Processes

Max. Marks: 15

Note: Attempt all questions. Assume suitable missing data, if any.

Q-1: Explain the working principle of Ultrasonic Machining (USM) with a neat diagram. Discuss the effect of process parameters such as abrasive grain size, slurry concentration, and frequency on material removal rate and surface finish. Also, explain the mechanism and types of tool wear occur in USM. (4)

Q-2: Describe the hydrostatic extrusion process with a neat sketch. Discuss its advantages and limitations compared to conventional extrusion processes. Also, highlight its practical industrial applications with suitable examples. (3)

Q-3: Explain the fundamental steps involved in the Additive Manufacturing process from CAD model to final product. Discuss the barriers and challenges in AM. Describe the working principle of material extrusion with a neat diagram, and explain why triangular elements are used in STL file preparation. (4)

Q-4: What is basic working of powder metallurgy process? Explain the complete process steps involved in powder metallurgy. Also, discuss the methods of powder production and the preparation and role of binders in the process. (4)

End of Question Paper